Controller perfboard - build in stages

Star Destroyer 1/2700, Arduino Nano ESP32

Short stages you can do one session at a time. Each ends with tests: do not move on until every box is ticked. Board 1 carries the Nano and Driver C (the 7 engines and the hangar bay); Board 2 carries Driver A and Driver B (the 10 hull zones, the bridge and the accent lights). Build Board 1 completely first (Stages 1-6), then Board 2 (Stages 7-8).

Boards: two 5 x 7 cm prototype boards, 18 x 24 holes. **Hole names:** columns **A-X** across the top from left to right, rows **1-18** with **row 1 at the bottom**, seen from the component side. "K16" means column K, row 16. If your board prints its own grid differently, mark these names on it in pencil first.

Every channel plugs in with a JST XH 2-pin connector. On each header, **pin 1 = +5 V** (red) and **pin 2 = OUT** (black). On Board 1 the 7 engine sockets are **right-angle (side-entry)** XH sockets with their openings facing the bottom edge, plus an eighth right-angle socket for the hangar bay in column T, opening facing the right-hand edge. Board 2 uses right-angle sockets too, split between its left, right and stern edges so each lead leaves on the side of the hull it serves. Make every LED lead with red on plug pin 1, and check one lead with the meter: its red wire must reach the socket's +5 V pin. Then any LED or cluster plugs into any channel, so you can test each channel as soon as it is wired and swap LEDs on and off freely.

Test lead: one XH lead with a 330 ohm resistor and an LED on the end (long leg to the resistor on the red wire) - the same as an engine lead on page 8. Plug it into a header to prove that channel. Tools: multimeter with a continuity beep, and a bench 5 V supply (your BreadVolt).

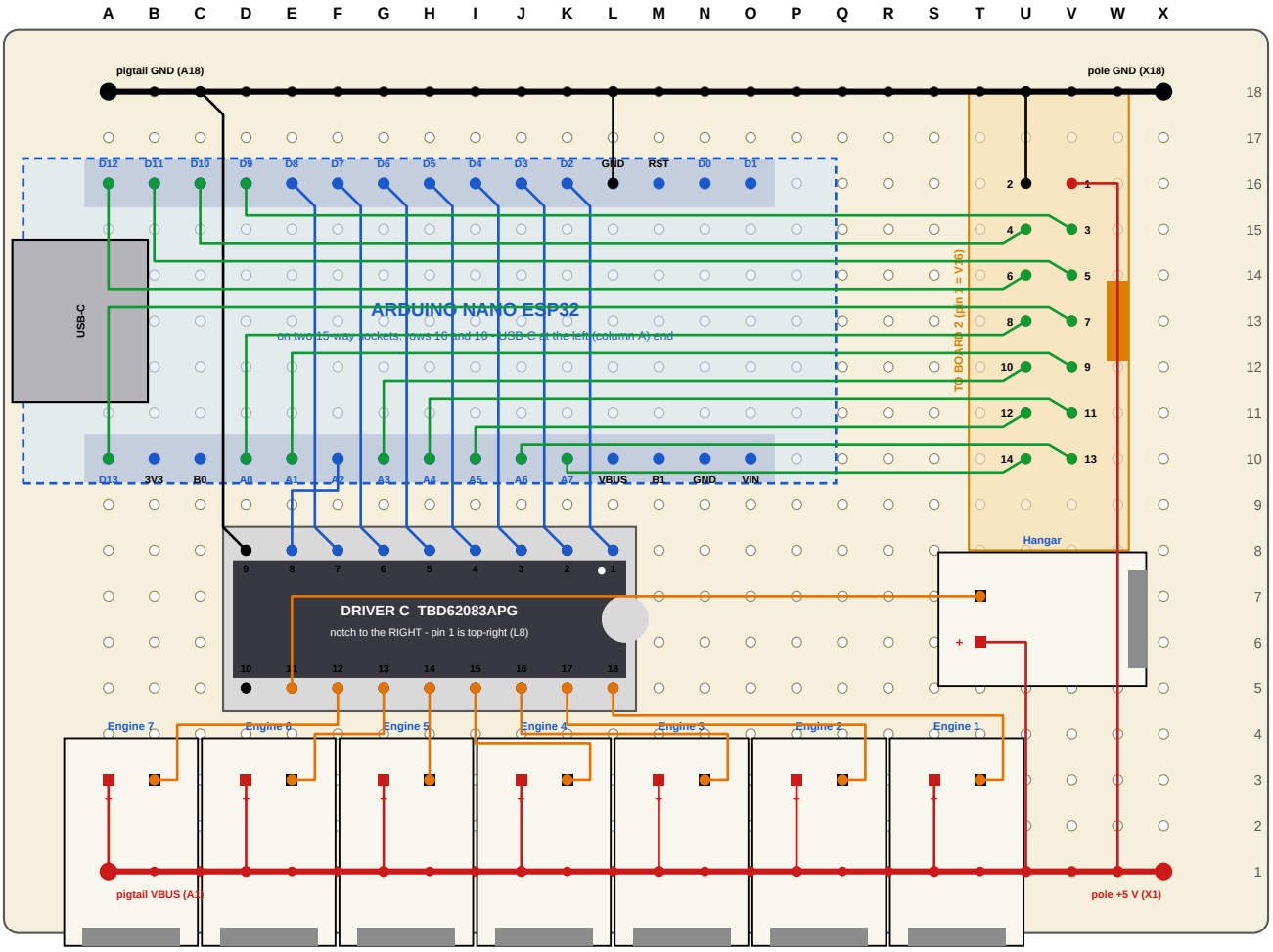
Lessons from the breadboard. Check ground with the beep test, never by voltage alone, because a floating pin can read 0 V. A poor ground shows up as the Nano resetting when many LEDs switch on at once.

Bus wires go in last. The two long bare power wires (GND along row 18, +5 V along row 1) are soldered only after every short wire is in place. Every other wire is thin **insulated** wire on the underside; insulated wires may cross each other, but never cross a bare bus wire.

Part (both boards)	Qty	Notes
5 x 7 cm prototype board, 18 x 24 holes, 2.54 mm pitch	2 + spare	Plain plated holes, no tracks
JST XH 2-pin right-angle socket (S2B-XH-A or compatible)	19 + spares	Board 1: 7 engines + hangar. Board 2: 5 left, 5 right, Accent
JST XH 2-pin vertical header (B2B-XH-A or compatible)	1 + spare	Board 2: Bridge (lead comes down from the upper hull)
JST XH 2-pin lead, pre-crimped, red and black, 150-200 mm	21 + spares	One per cluster, one for the accent lights, plus the test lead
18-pin DIP socket / TBD62083APG	3 / 3	Plus spares
15-way female socket strip	2	For the Nano
14-way shrouded box header + 14-way IDC ribbon cable	2 + 1	Board-to-board harness
Thin insulated wire (28-30 AWG) and 0.6 mm tinned bare wire	-	Short wires and the two buses
Accent lights: stripboard, 6 x 1 k resistors, 6 pre-wired SMD LEDs (no built-in resistor)	2 boards / 6 / 6	Two junction boards, 3 x 11 holes (page 9)

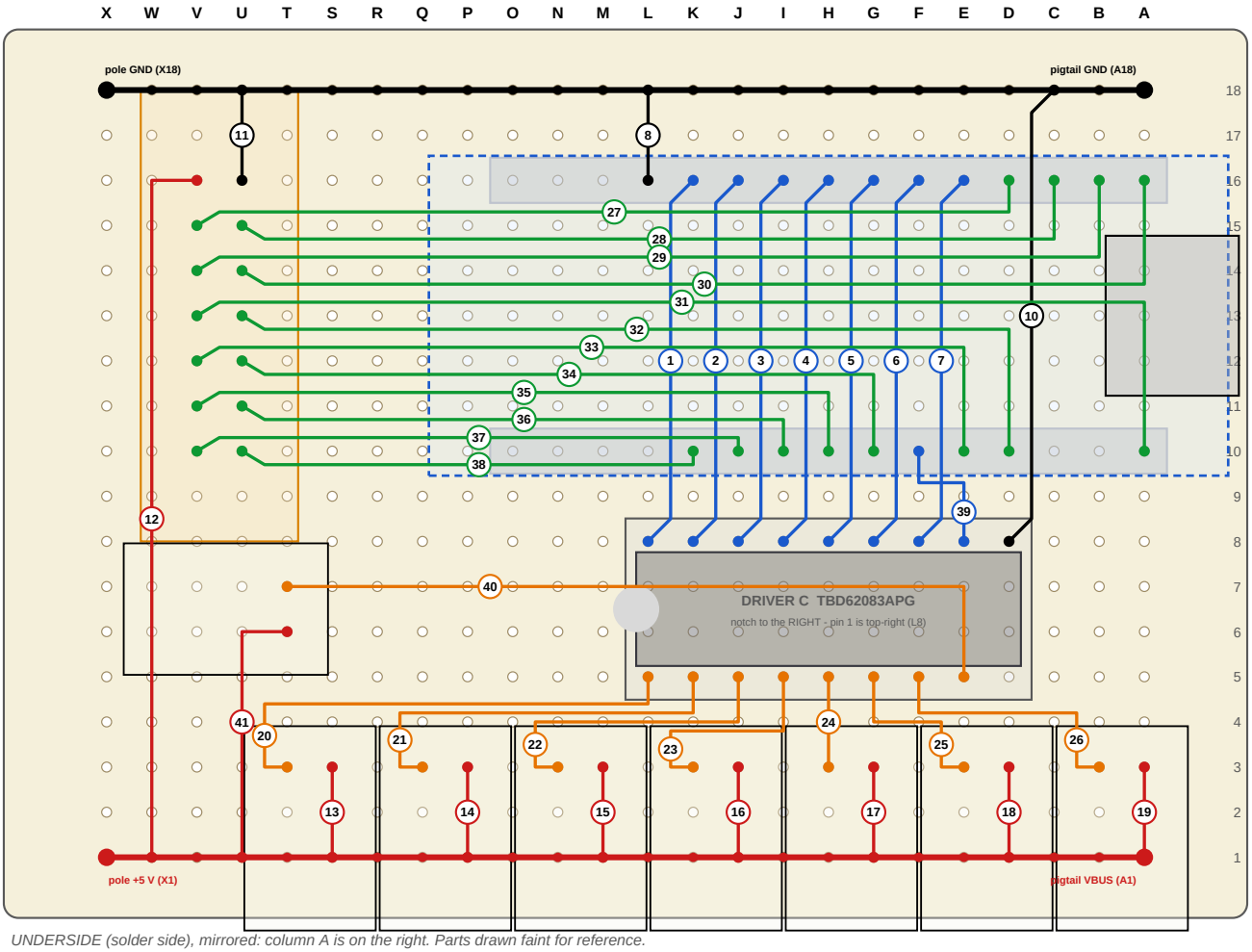
Board 1 - component side

Column A on the left. Nano on sockets in rows 16 and 10 (USB-C at the left edge). Driver C on its side in rows 8 and 5: **notch to the right, pin 1 at L8**. Engine sockets (right-angle, opening toward the bottom edge) along row 3 - Engine 7 on the left (A-B) to Engine 1 on the right (S-T); +5 V pins in A, D, G, J, M, P, S. Box header at U-V, pin 1 at V16. **Accent socket** (right-angle XH) in column T, between Driver C and the box header, its opening facing the right-hand edge: pin 1 (+5 V) at T6, pin 2 (OUT) at T7. This keeps the plug clear of the thick +5 V wire from the box header down column W.



Board 1 - underside

Turned over like a page, so column A is on the right. Each wire has its number from the table.



UNDERSIDE (solder side), mirrored: column A is on the right. Parts drawn faint for reference.

Board 1 - wires to fit (tick each one)

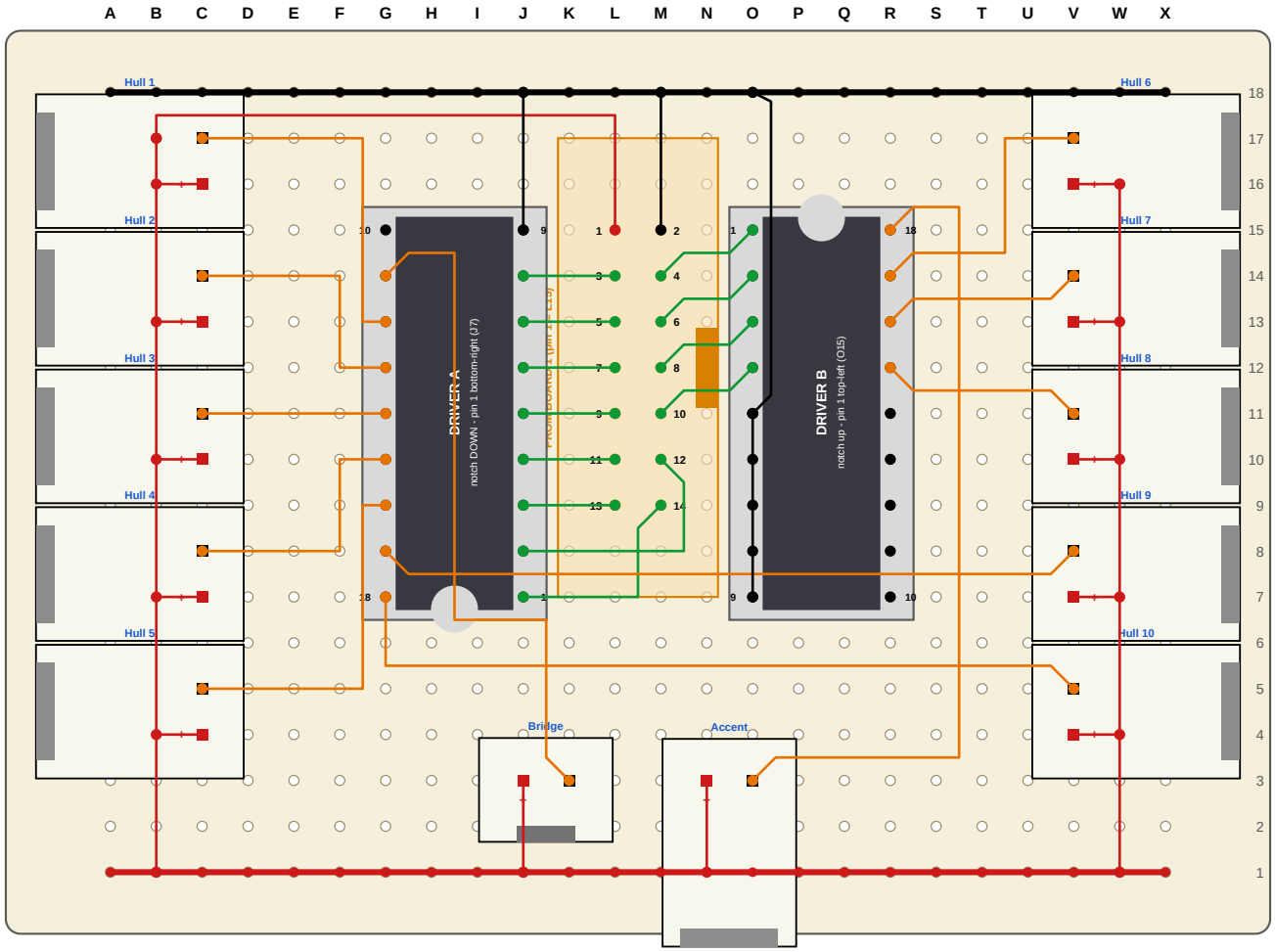
Buses (bare wire, laid last): **GND along row 18** and **+5 V along row 1**, A to X. Pole wires: X1 (+5 V) and X18 (GND). USB-C power pigtail: A1 (VBUS) and A18 (GND). Driver C pin 10 (D5, COMMON) stays unconnected; Nano B0, B1, 3V3, VBUS and VIN are not used. There is **no wire 9**: Driver C input 8 is the hangar bay channel (wires 39-41).

Tick	#	Colour	What it joins	From	To
[ ]	1	Blue	Nano D2 to Driver C pin 1 (engine 1)	K16	L8
[ ]	2	Blue	Nano D3 to Driver C pin 2 (engine 2)	J16	K8
[ ]	3	Blue	Nano D4 to Driver C pin 3 (engine 3)	I16	J8
[ ]	4	Blue	Nano D5 to Driver C pin 4 (engine 4)	H16	I8
[ ]	5	Blue	Nano D6 to Driver C pin 5 (engine 5)	G16	H8
[ ]	6	Blue	Nano D7 to Driver C pin 6 (engine 6)	F16	G8
[ ]	7	Blue	Nano D8 to Driver C pin 7 (engine 7)	E16	F8
[ ]	8	Black	Nano GND to the GND bus	L16	L18
[ ]	10	Black	Driver C pin 9 (GND) to the GND bus	D8	C18
[ ]	11	Black	Box header pin 2 (GND) to the GND bus	U16	U18
[ ]	12	Red	Box header pin 1 (+5 V) to the +5 V bus	V16	W1
[ ]	13	Red	Engine 1 socket pin 1 (+5 V) to the +5 V bus	S3	S1
[ ]	14	Red	Engine 2 socket pin 1 (+5 V) to the +5 V bus	P3	P1
[ ]	15	Red	Engine 3 socket pin 1 (+5 V) to the +5 V bus	M3	M1
[ ]	16	Red	Engine 4 socket pin 1 (+5 V) to the +5 V bus	J3	J1
[ ]	17	Red	Engine 5 socket pin 1 (+5 V) to the +5 V bus	G3	G1
[ ]	18	Red	Engine 6 socket pin 1 (+5 V) to the +5 V bus	D3	D1
[ ]	19	Red	Engine 7 socket pin 1 (+5 V) to the +5 V bus	A3	A1
[ ]	20	Orange	Driver C pin 18 to Engine 1 socket pin 2 (OUT)	L5	T3
[ ]	21	Orange	Driver C pin 17 to Engine 2 socket pin 2 (OUT)	K5	Q3
[ ]	22	Orange	Driver C pin 16 to Engine 3 socket pin 2 (OUT)	J5	N3
[ ]	23	Orange	Driver C pin 15 to Engine 4 socket pin 2 (OUT)	I5	K3
[ ]	24	Orange	Driver C pin 14 to Engine 5 socket pin 2 (OUT)	H5	H3
[ ]	25	Orange	Driver C pin 13 to Engine 6 socket pin 2 (OUT)	G5	E3
[ ]	26	Orange	Driver C pin 12 to Engine 7 socket pin 2 (OUT)	F5	B3
[ ]	27	Green	Nano D9 to box header pin 3	D16	V15
[ ]	28	Green	Nano D10 to box header pin 4	C16	U15
[ ]	29	Green	Nano D11 to box header pin 5	B16	V14
[ ]	30	Green	Nano D12 to box header pin 6	A16	U14
[ ]	31	Green	Nano D13 to box header pin 7	A10	V13
[ ]	32	Green	Nano A0 to box header pin 8	D10	U13
[ ]	33	Green	Nano A1 to box header pin 9	E10	V12
[ ]	34	Green	Nano A3 to box header pin 10	G10	U12
[ ]	35	Green	Nano A4 to box header pin 11	H10	V11
[ ]	36	Green	Nano A5 to box header pin 12	I10	U11
[ ]	37	Green	Nano A6 to box header pin 13	J10	V10
[ ]	38	Green	Nano A7 to box header pin 14	K10	U10
[ ]	39	Blue	Nano A2 to Driver C pin 8 (hangar bay)	F10	E8
[ ]	40	Orange	Driver C pin 11 to Hangar socket pin 2 (OUT)	E5	T7
[ ]	41	Red	Hangar socket pin 1 (+5 V) to the +5 V bus at U1	T6	U1

Board 2 - component side

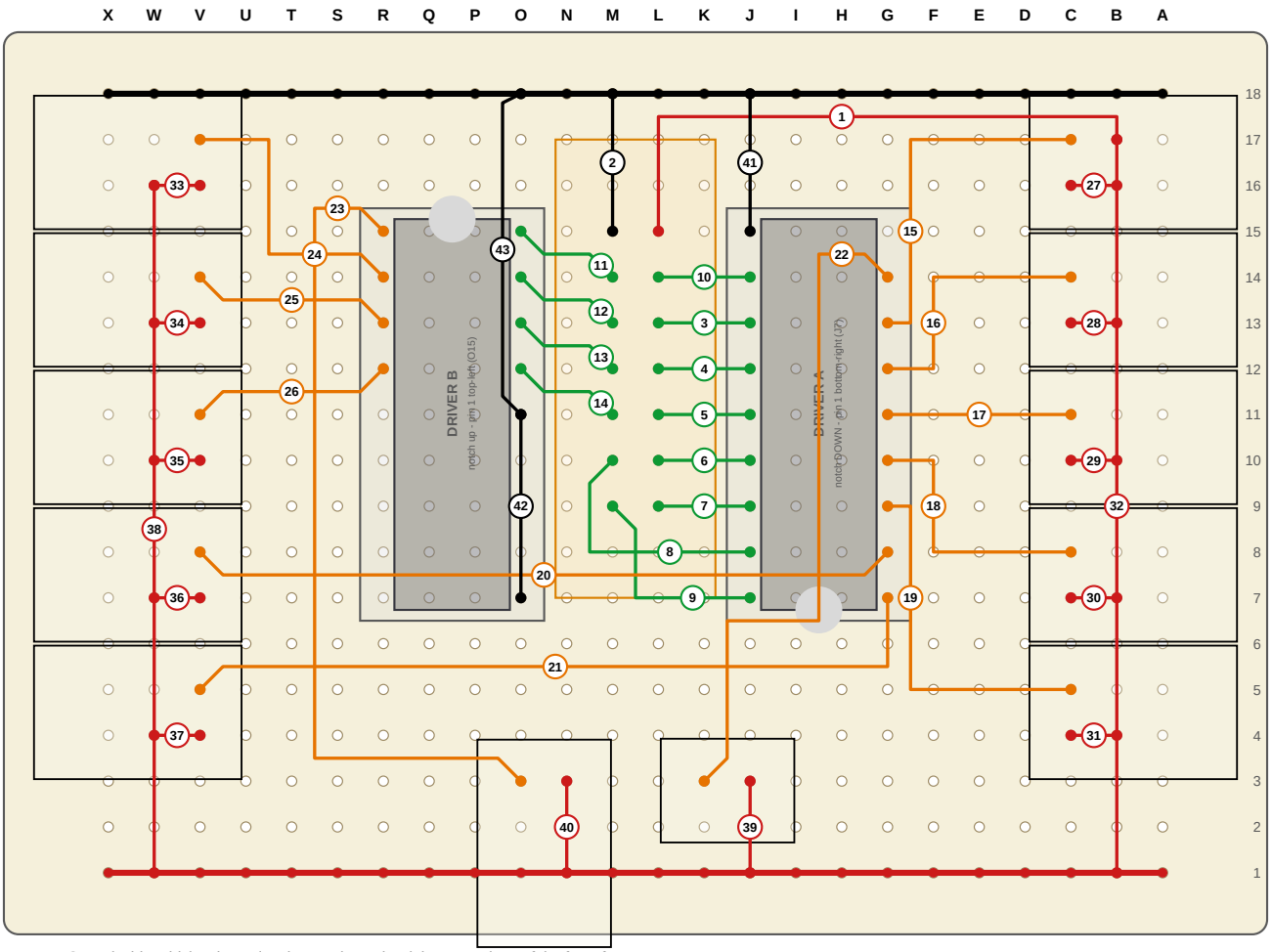
Board 2 sockets are grouped by where the leads go. Mount Board 2 with column A toward the **left** side of the hull, column X toward the **right** side and the row 1 edge toward the **stern and the support pole**. Left edge: Hull 1-5, pins in column C (+5 V at C16, C13, C10, C7, C4; OUT at C17, C14, C11, C8, C5), openings facing left. Right edge: Hull 6-10, pins in column V (+5 V at V16, V13, V10, V7, V4; OUT at V17, V14, V11, V8, V5), openings facing right. Stern edge: Bridge J3/K3 on an **upright** header (plug from above, as its lead comes down from the upper hull) and Accent N3/O3 on a right-angle socket facing the bottom edge (+5 V on the left pin of both).

The driver block sits in the middle of the board, the same distance from the left and right socket columns and from the stern sockets and the top bus. Driver A in columns G-J and Driver B in columns O-R, rows 15-7. **Driver A is turned round: notch at the BOTTOM, pin 1 bottom-right at J7**, so its inputs face the box header and its outputs face the left-hand sockets. Driver B is notch at the top, pin 1 at O15. The box header stands upright between the two drivers in columns L-M, rows 15-9, **pin 1 at L15**.



Socket	Faces	+5 V pin	OUT pin	Driver output	Nano pin
Hull 1	Left side of hull	C16	C17	A pin 12	D11
Hull 2	Left side of hull	C13	C14	A pin 13	D13
Hull 3	Left side of hull	C10	C11	A pin 14	A1
Hull 4	Left side of hull	C7	C8	A pin 15	A4
Hull 5	Left side of hull	C4	C5	A pin 16	A6
Hull 6	Right side of hull	V16	V17	B pin 17	D12
Hull 7	Right side of hull	V13	V14	B pin 16	A0
Hull 8	Right side of hull	V10	V11	B pin 15	A3
Hull 9	Right side of hull	V7	V8	A pin 17	A5
Hull 10	Right side of hull	V4	V5	A pin 18	A7
Bridge	Upright (plug from above)	J3	K3	A pin 11	D9
Accent	Stern / pole	N3	O3	B pin 18	D10

Board 2 - underside



UNDERSIDE (solder side), mirrored: column A is on the right. Parts drawn faint for reference.

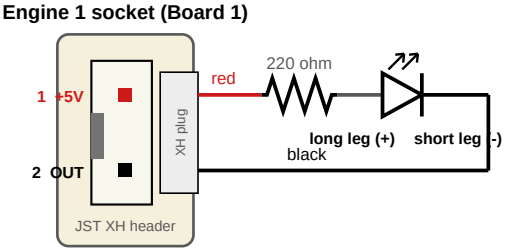
Board 2 - wires to fit (tick each one)

Buses (bare wire, laid last): **GND along row 18** and **+5 V along row 1**, A to X. Two red risers carry +5 V up the sides: column B (wire 32) for the left sockets and column W (wire 38) for the right sockets. Driver A pin 10 (G15) and Driver B pins 10-14 (R7-R11) stay unconnected.

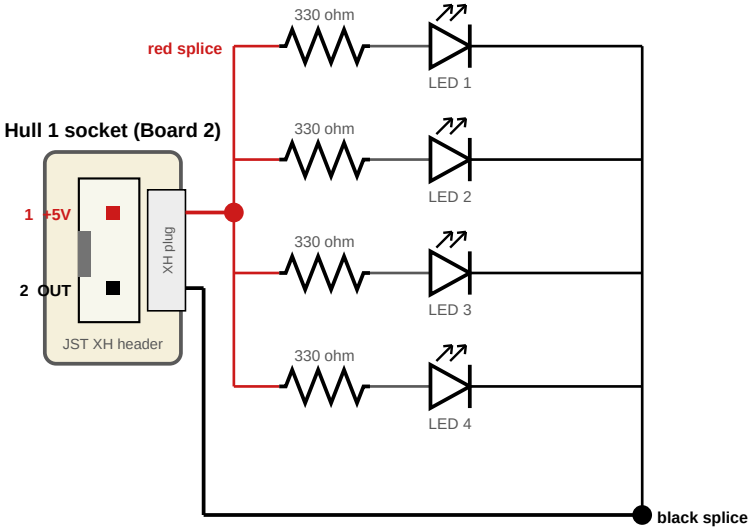
Tick	#	Colour	What it joins	From	To
[ ]	1	Red	Box header pin 1 (+5 V) to the left +5 V riser	L15	B17
[ ]	2	Black	Box header pin 2 (GND) to the GND bus	M15	M18
[ ]	3	Green	Box header pin 5 (D11) to Driver A pin 7	L13	J13
[ ]	4	Green	Box header pin 7 (D13) to Driver A pin 6	L12	J12
[ ]	5	Green	Box header pin 9 (A1) to Driver A pin 5	L11	J11
[ ]	6	Green	Box header pin 11 (A4) to Driver A pin 4	L10	J10
[ ]	7	Green	Box header pin 13 (A6) to Driver A pin 3	L9	J9
[ ]	8	Green	Box header pin 12 (A5) to Driver A pin 2	M10	J8
[ ]	9	Green	Box header pin 14 (A7) to Driver A pin 1	M9	J7
[ ]	10	Green	Box header pin 3 (D9) to Driver A pin 8	L14	J14
[ ]	11	Green	Box header pin 4 (D10) to Driver B pin 1	M14	O15
[ ]	12	Green	Box header pin 6 (D12) to Driver B pin 2	M13	O14
[ ]	13	Green	Box header pin 8 (A0) to Driver B pin 3	M12	O13
[ ]	14	Green	Box header pin 10 (A3) to Driver B pin 4	M11	O12
[ ]	15	Orange	Driver A pin 12 to Hull 1 socket pin 2 (OUT)	G13	C17
[ ]	16	Orange	Driver A pin 13 to Hull 2 socket pin 2 (OUT)	G12	C14
[ ]	17	Orange	Driver A pin 14 to Hull 3 socket pin 2 (OUT)	G11	C11
[ ]	18	Orange	Driver A pin 15 to Hull 4 socket pin 2 (OUT)	G10	C8
[ ]	19	Orange	Driver A pin 16 to Hull 5 socket pin 2 (OUT)	G9	C5
[ ]	20	Orange	Driver A pin 17 to Hull 9 socket pin 2 (OUT)	G8	V8
[ ]	21	Orange	Driver A pin 18 to Hull 10 socket pin 2 (OUT)	G7	V5
[ ]	22	Orange	Driver A pin 11 to Bridge socket pin 2 (OUT)	G14	K3
[ ]	23	Orange	Driver B pin 18 to Accent socket pin 2 (OUT)	R15	O3
[ ]	24	Orange	Driver B pin 17 to Hull 6 socket pin 2 (OUT)	R14	V17
[ ]	25	Orange	Driver B pin 16 to Hull 7 socket pin 2 (OUT)	R13	V14
[ ]	26	Orange	Driver B pin 15 to Hull 8 socket pin 2 (OUT)	R12	V11
[ ]	27	Red	Hull 1 socket pin 1 (+5 V) to the left riser	C16	B16
[ ]	28	Red	Hull 2 socket pin 1 (+5 V) to the left riser	C13	B13
[ ]	29	Red	Hull 3 socket pin 1 (+5 V) to the left riser	C10	B10
[ ]	30	Red	Hull 4 socket pin 1 (+5 V) to the left riser	C7	B7
[ ]	31	Red	Hull 5 socket pin 1 (+5 V) to the left riser	C4	B4
[ ]	32	Red	Left +5 V riser down column B to the +5 V bus	B17	B1
[ ]	33	Red	Hull 6 socket pin 1 (+5 V) to the right riser	V16	W16
[ ]	34	Red	Hull 7 socket pin 1 (+5 V) to the right riser	V13	W13
[ ]	35	Red	Hull 8 socket pin 1 (+5 V) to the right riser	V10	W10
[ ]	36	Red	Hull 9 socket pin 1 (+5 V) to the right riser	V7	W7
[ ]	37	Red	Hull 10 socket pin 1 (+5 V) to the right riser	V4	W4
[ ]	38	Red	Right +5 V riser down column W to the +5 V bus	W16	W1
[ ]	39	Red	Bridge socket pin 1 (+5 V) to the +5 V bus	J3	J1
[ ]	40	Red	Accent socket pin 1 (+5 V) to the +5 V bus	N3	N1
[ ]	41	Black	Driver A pin 9 (GND) to the GND bus	J15	J18
[ ]	42	Black	Driver B unused inputs 5-8 joined to pin 9	O11	O7
[ ]	43	Black	Driver B pins 5-9 (GND) to the GND bus	O11	O18

How a channel is wired

Every channel is a 2-pin JST XH header: **pin 1 = +5 V** (from the +5 V bus) and **pin 2 = OUT** (from a driver output). The LED lead ends in an XH plug: red wire to pin 1, black to pin 2. When the sketch turns the channel on, the driver connects OUT to ground and current flows: pin 1, resistor, LED long leg, LED short leg, pin 2, driver, GND. The resistor sets the current, so every LED has its own.



One LED and its 220 ohm resistor, no splice: red to plug pin 1, black to plug pin 2. Engines 2-7 are the same; the test lead is this with 330 ohm.

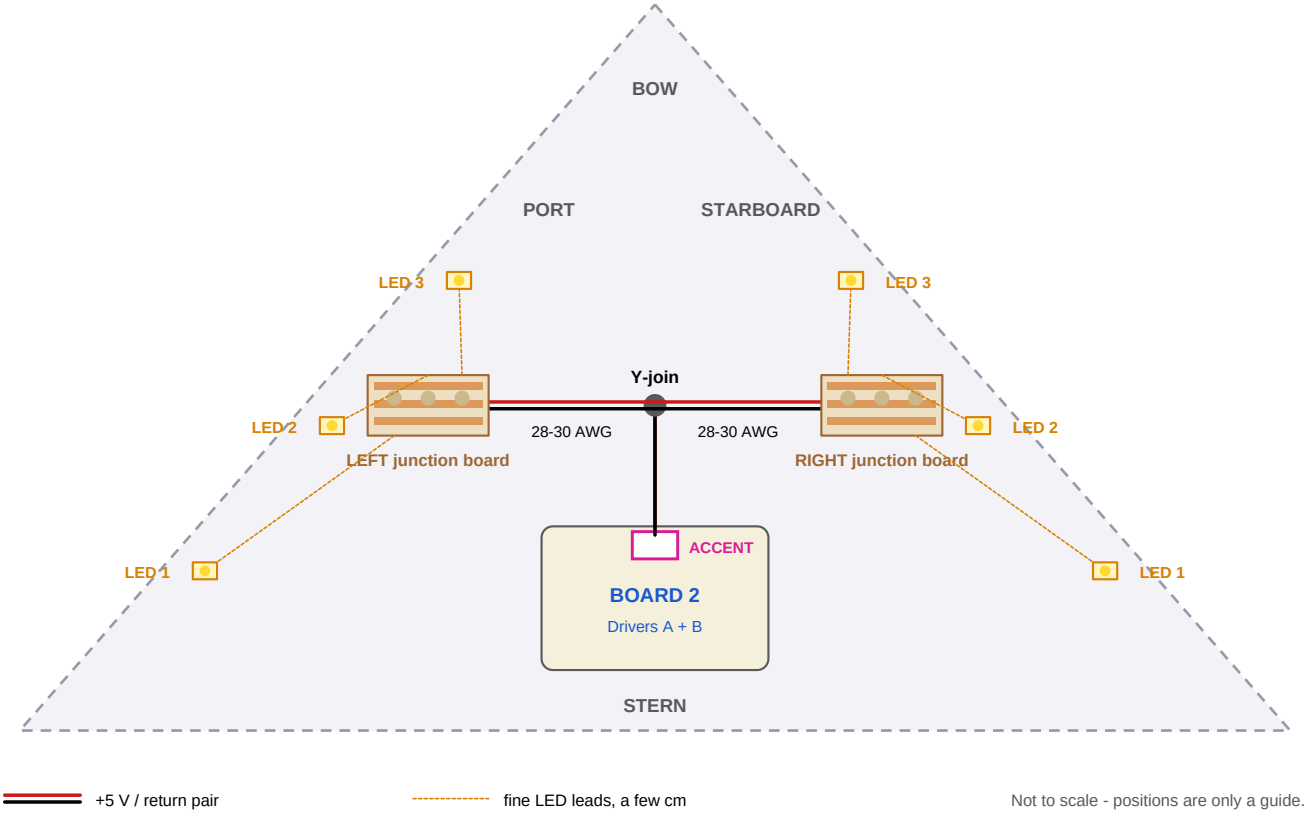


Each LED gets its own resistor on its long (+) leg. All 4 red leads join the red wire to plug pin 1 (+5V); all 4 black leads join the black wire to plug pin 2 (OUT). Solder every splice and cover it in heat-shrink. The bridge and the hangar bay are the same, 4 LEDs each.

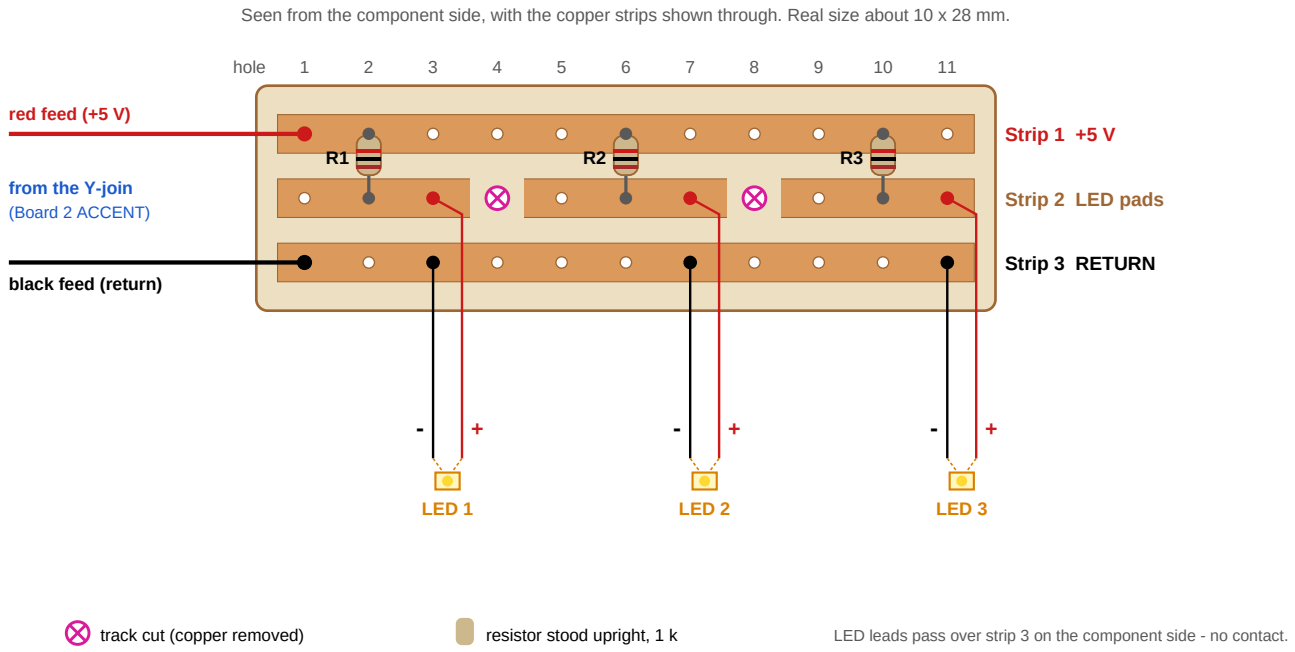
Engines 1-7 on Board 1 are the left drawing. The hull zones, bridge and hangar bay on Board 2 are the right-hand drawing. If an LED is in backwards, its channel stays dark but nothing is damaged: turn it round.

Channel 20 - the accent lights

Six SMD LEDs, three along each side of the hull, on one channel: Nano **D10** drives Driver B input 1 on Board 2, and its output (pin 18) goes to the **Accent** socket on Board 2's stern edge (N3/O3). One lead from that header splits at a Y-join into two red/black pairs (28-30 AWG), one to a small **junction stripboard** on each side. Each junction board carries three 1 k resistors, and the LEDs' fine pre-wired leads run only a few centimetres from the board. All six fade up together as a steady glow, 14 s into the show.



Each junction board is 3 strips by 11 holes. Strip 1 is +5 V, strip 3 is the return, and strip 2 is cut at holes 4 and 8 to make three separate LED pads:



Before fitting each junction board: beep test that strip 1 does not beep to strip 3 and the three pads are separate; then power it from the bench supply (**+5v** to the red feed, **negative** to the black feed) and check all three LEDs light. Cover the copper side with Kapton tape, glue it in, and tack the fine leads down with a slack loop at each LED. Start with 1 k resistors; use 470 R if the LEDs look too dim.

Stage 1 - Board 1: mark out and dry-fit

Goal: Know exactly where everything goes, before any solder.

Do

- [ ] Mark the hole names and the positions from the Board 1 drawing in pencil.
- [ ] Without soldering, place the two 15-way Nano sockets (rows 16 and 10, columns A-O), the DIP socket (rows 8 and 5, columns D-L, notch to the right), the box header (columns U-V, rows 16-10, pin 1 at V16) and the 7 right-angle XH sockets along row 3, openings facing the bottom edge, and the right-angle Hangar socket at T6-T7 with its opening facing the right-hand edge.
- [ ] Fit the Nano: its USB-C end is at the left edge, over column A.

Tests - all must pass before the next stage

- [ ] Every part sits flat on the marked holes; no XH header touches its neighbour.
- [ ] The Nano clears the box header and the DIP socket, and a USB-C plug fits.

Stage 2 - Board 1: sockets and headers

Goal: The Nano sockets, the Driver C socket, the box header, the 7 engine sockets and the Hangar socket are soldered flat, with no wires yet.

Do

- [ ] Solder each part: tack one pin, check it is flat and square, then do the rest. Leave the chip and the Nano out.
- [ ] XH sockets: pin 1 (+5 V) in columns A, D, G, J, M, P, S and pin 2 (OUT) in B, E, H, K, N, Q, T, all on row 3.
- [ ] Hangar socket (right-angle): pin 1 (+5 V) at T6, pin 2 (OUT) at T7, opening facing the right-hand edge. Check a plug goes in freely before soldering.

Tests - all must pass before the next stage

- [ ] Everything sits flat on the board, with no gap or tilt.
- [ ] Beep test: no two neighbouring pins beep to each other (no solder bridges).

Stage 3 - Board 1: short power and ground wires

Goal: Every ground and +5 V connection is wired to its bus row; the two bus wires are NOT laid yet.

Do

- [ ] Ground (wires 8, 10, 11 - there is no wire 9): Nano GND L16 to L18; Driver C pin 9 (D8) up between columns C and D to C18; box header pin 2 U16 to U18. Driver C pin 8 (E8) is **not** grounded: it is the hangar bay input.
- [ ] +5 V (wires 12-19 and 41): box header pin 1 V16 across to W16 and down column W to W1; each engine socket pin 1 straight down to row 1 (A3 to A1, D3 to D1 ... S3 to S1); Hangar socket pin 1 (T6) one hole right to U6, then down column U to U1.
- [ ] End each wire at its bus-row hole on the underside, with a short bare tip ready for the bus wire.

Tests - all must pass before the next stage

- [ ] Beep test: each wire beeps from end to end.
- [ ] Beep test between any +5 V wire and any GND wire: **must NOT beep**.

Stage 4 - Board 1: bus wires last, then power up

Goal: The two bus wires are in, the board powers up, and the Nano runs.

Do

- [ ] Lay a straight bare tinned wire (about 0.6 mm) along row 18 (GND) and row 1 (+5 V), A to X, and solder it at every hole.
- [ ] Solder the bench power leads (later the pole wires, through the fuse) to X1 (+5 V) and X18 (GND). Use plenty of solder; the ground joint must be strong.
- [ ] Make the USB-C power pigtail from A1 (VBUS) and A18 (GND). Never plug it in while the Nano is connected to your computer.

Tests - all must pass before the next stage

- [ ] Beep test: along the whole +5 V bus and the whole GND bus; the two buses must **NOT** beep to each other.
- [ ] Bench supply 5 V: 5 V between the buses at both ends, and the supply current is close to zero.
- [ ] Every engine socket and the Hangar socket: 5 V between pin 1 and the GND bus.
- [ ] With only the computer cable, upload Blink: the small orange LED flashes. Then, powered from the pigtail alone, the Nano starts up (RGB LED lit). Unplug it again.

Stage 5 - Board 1: signal wires

Goal: Driver C and the box header are wired to the Nano, and the chip is fitted.

Do

- [ ] **Green wires first.** Wires 27-38: the 12 green wires from the Nano to the box header, one at a time from the table. Several of them solder to the Nano's bottom row of pins (row 10), so do them all before the blue wires, which pass between those pins.
- [ ] **Then the blue wires.** Wires 1-7: Nano D2 (K16) to Driver C pin 1 (L8), D3 (J16) to pin 2 (K8), and so on to D8 (E16) to pin 7 (F8). Each runs between two columns, so it never passes over another pin. They are only soldered at row 16 and row 8, never at row 10. If you need to re-solder a row 10 pin later, lay a strip of Kapton tape over the blue wires first.
- [ ] Wire 39: Nano A2 (F10) one hole down, across to column E and into Driver C pin 8 (E8). It crosses the D8 wire; both are insulated, so that is fine.
- [ ] Fit the chip: **notch to the right**, pin 1 at L8.

Tests - all must pass before the next stage

- [ ] Beep test: each wire end to end, and no wire beeps to its neighbour.
- [ ] No chip leg is bent under the body, and the notch is to the right.

Stage 6 - Board 1: outputs and the engines

Goal: All 7 engine channels and the hangar bay channel switch a real load. This proves the whole Board 1 path.

Do

- [ ] Wires 20-26: Driver C pin 18 (L5) to Engine 1 pin 2 (T3), pin 17 (K5) to Engine 2 (Q3), and so on to pin 12 (F5) to Engine 7 (B3), as drawn (no crossings). Pin 14 (H5) runs straight down to Engine 5 (H3).
- [ ] Wire 40: Driver C pin 11 (E5) up to row 7 (between the chip's pin rows), then right along row 7, under the chip, to the Hangar socket pin 2 (T7).

Tests - all must pass before the next stage

- [ ] Beep test: each output to its header pin 2; no output beeps to another output or to GND.
- [ ] **Take the Nano out of its sockets first** (its pins are 3.3 V only and +5 V would damage them). Plug the test lead into Engine 1. Hold a wire from +5 V to Driver C pin 1 for a moment: the LED lights. Repeat for Engines 2-7 with pins 2-7.
- [ ] Still with the Nano out: plug the test lead into the Hangar socket and hold +5 V on Driver C pin 8 (E8) for a moment: the LED lights. Refit the Nano.

- [ ] Plug in all 7 engine LEDs and upload **engine_test_nano.ino**: one at a time, all together, fade each, production look.
- [ ] **All together test:** in step 2 all 7 light with no Nano reset. A reset means a poor ground joint.

Stage 7 - Board 2: sockets and power

Goal: Board 2 has its parts, its power and ground wiring and its buses, and gets power over the ribbon cable.

Do

- [ ] Solder the two DIP sockets: **Driver A notch DOWN** (columns G-J, rows 15-7, pin 1 at J7) and **Driver B notch UP** (columns O-R, pin 1 at O15). Solder the box header upright between them in columns L-M, rows 15-9, pin 1 at L15. Dry-fit all three first: the box header only just fits between the two DIP sockets.
- [ ] Solder the 12 XH connectors: Hull 1-5 down the left edge (pins in column C, openings facing left), Hull 6-10 down the right edge (pins in column V, openings facing right), and along the stern edge the Bridge as an **upright** header (J3/K3, lock ramp toward the bottom edge) and the Accent as a right-angle socket (N3/O3, opening facing down). +5 V is the lower pin on the side sockets and the left pin on the stern pair. Check each plug goes in freely before soldering.
- [ ] Power and ground wires: box header pin 1 L15 up and left along the lane just below row 18 to B17 (wire 1, thick wire - it carries all of Board 2's current); box header pin 2 M15 to M18 (wire 2); left sockets' +5 V pins one hole left to column B (wires 27-31); right sockets' +5 V pins one hole right to column W (wires 33-37); stern sockets' +5 V pins straight down to row 1 (wires 39-40); Driver A pin 9 J15 to J18 (wire 41); Driver B pins 5-9 (O11 to O7) joined and run up to O18 (wires 42-43).
- [ ] Risers and buses last: bare or thick red wire up column B from B1 to B17 (wire 32) and up column W from W1 to W16 (wire 38), soldered at every socket tap; then bare wire along row 18 (GND) and row 1 (+5 V), A to X.

Tests - all must pass before the next stage

- [ ] Beep tests: no bridges; each wire end to end; the +5 V bus does **not** beep to the GND bus.
- [ ] Plug the ribbon cable between the two boards (red stripe to pin 1 at both ends) and power Board 1: Board 2 reads 5 V between its buses, on both risers and on every XH socket's pin 1.

Stage 8 - Board 2: signals, outputs and all 20 channels

Goal: Every channel on both boards works from the Nano.

Do

- [ ] Wires 3-14: the 12 green wires from the box header to the driver inputs. The L-column pins run straight across to Driver A's inputs in column J; the M-column pins step up one row into Driver B's inputs in column O (two of them, M9 and M10, drop down to J7 and J8).
- [ ] Wires 15-22: Driver A outputs (column G) to the left sockets, Hull 9 and Hull 10 on the right edge, and the Bridge on the stern edge. Wire 20 (Hull 9) runs right between the drivers' bottom two pin rows, wire 21 (Hull 10) runs right below both chips, and wire 22 (Bridge) runs down under Driver A between its pin rows. Wires 23-26: Driver B outputs (column R) to the Accent socket and Hull 6-8 on the right edge.
- [ ] Fit the chips: **Driver A notch DOWN**, Driver B notch UP.

Tests - all must pass before the next stage

- [ ] Beep test through the ribbon cable: each Nano pin on Board 1 to its driver input on Board 2 (for example Nano D11 to Driver A pin 7 at J13, and Nano D10 to Driver B pin 1 at O15).
- [ ] Upload **pin_test_nano.ino** and plug the test lead into each header in turn: every channel lights in its turn: engines 1-7, bridge, hangar (Board 1), Hull 1-10 in socket order, then the accent lights.
- [ ] Nothing lights that should be off, and the Nano does not reset at any point.

Stage 9 - Full system and soak

Goal: The finished electronics are ready to go into the hull.

Do

- [ ] Plug in all 19 LED clusters (Stage 3 of the build guide) and the accent lead with both junction boards (Board 2's Accent socket), then upload the main sketch.
- [ ] Run the show from the Bluefruit Connect app, ON then OFF.

Tests - all must pass before the next stage

- [ ] All 20 channels behave correctly in the show; engines breathe and shimmer, and the accent lights fade up to a steady glow.
- [ ] Run for at least 30 minutes with no resets and no hot parts (feel the chips and resistors).
- [ ] Check the supply current stays within your fuse rating. Finish all of this before closing the hull.